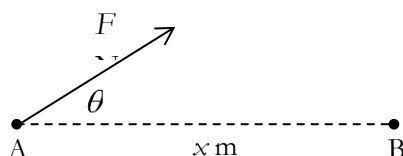


Kinetic energy

The kinetic energy of a body of mass m and speed v is defined to be $\frac{1}{2}mv^2$.

Work done by a force

Work is done **by** a force **on** a body. If a particle moves along a straight line segment AB of length x metres under the action of some force with constant magnitude F N and direction θ , we define the work done by that force to be $Fx \cos \theta$ joules (J). Note that work done is a **signed scalar** quantity:

- If the component of the force parallel to AB is *with* the direction of motion, the work done is positive;
- If the component of the force parallel to AB is *against* the direction of motion, the work done is negative.

The Work-Energy Principle:

(energy before) + (work done by "driving forces") – (work done against resistance forces) = (energy after)

1. A cricket ball of mass 400 g moving at 3 ms^{-1} and a golf ball of mass 100 g have equal kinetic energies. Find the speed at which the golf ball is moving.
2. A body of mass 4 kg has -32 J of work done on it by resistance forces. If initially it had a speed of 5 ms^{-1} , find its final speed.
3. A sledge is pulled for 5 m across a smooth ice surface by a force of 3 N. Find the work done on the sledge by this force.
4. A constant force F N pulls a body of mass 0.5 kg in a straight line across a smooth horizontal surface. The body passes through a point A with a speed of 3 ms^{-1} and then through a point B with a speed of 5 ms^{-1} , B being 20 m from A. Neglect any resistances to motion.
 - a. Let W be the work done by the pulling force. Explain why W satisfies the equation $\left(\frac{1}{2} \times \frac{1}{2} \times 3^2\right) + W = \left(\frac{1}{2} \times \frac{1}{2} \times 5^2\right)$, and solve it to find W .
 - b. Find the magnitude of the force, by using the equation

$$(\text{work done}) = (\text{force}) \times \left(\frac{\text{distance moved}}{\text{in direction of force}} \right)$$
5. A body of mass 10 kg, moving under the action of a single constant force, accelerates from 2 ms^{-1} to 5 ms^{-1} .
 - a. Write an equation using the Work-Energy Principle, and solve it to find the work done by the force.
 - b. Given that the magnitude of the force is 7 N, write an equation and solve it to find the distance d m travelled by the particle during the acceleration.

6. A body of mass 5 kg slides over a rough horizontal surface. In sliding 5 m the speed of the body decreases from 8 ms^{-1} to 6 ms^{-1} .
 - a. Let W be the work done by the frictional forces. Write down an equation satisfied by W and solve it to find the work done *by* the frictional forces.
 - b. State the work done *against* the frictional forces.
7. A particle of mass 100 g moves in a straight line across a horizontal surface against resistances of constant magnitude. The particle passes through a point A with a speed of 7 ms^{-1} and then through a point B with a speed of 3 ms^{-1} , B being 4 m from A. Find the magnitude of the resistances.
8. A car of mass 1200 kg is travelling in a straight line along a horizontal road when it performs an emergency stop. During this procedure there is a constant braking force of 7800 N. Given that the car travels a distance of 38 m before stopping, find the speed with which the car was originally travelling.